CS4483B / 9541 Game Design

Deadlight: Survival After Dark

Deliverable 2 Mid-Project Report

Group 16

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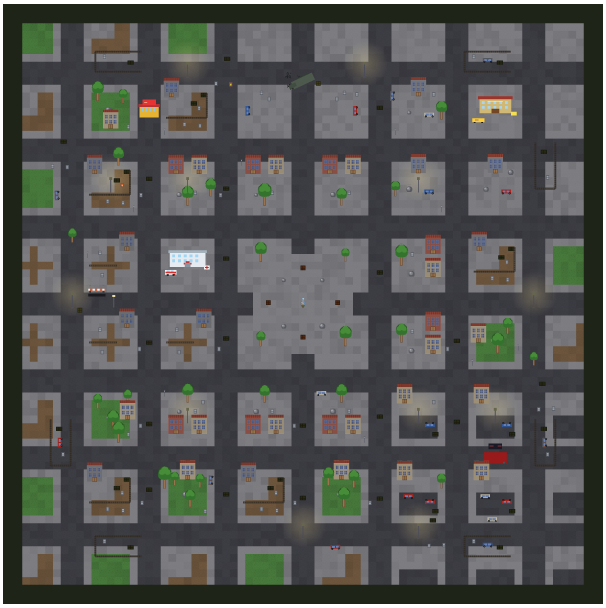
Submission Scope:	Deliverable 2 playable prototype covering Levels 1-2
Playable Length:	Six objective nights across two campaign spaces
Project Type:	Top-down zombie survival prototype in Unity
Report Focus:	Level design, guidance, narrative, systems, progression, and iteration

1 Project Overview

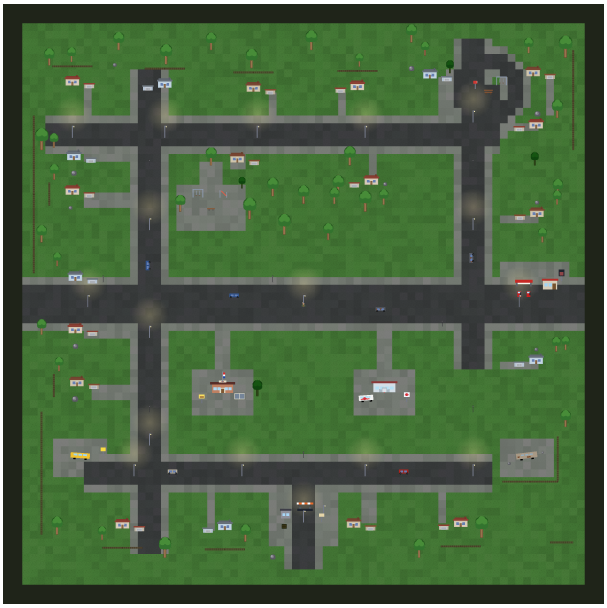
Deliverable 2 presents a playable prototype of *Deadlight* in a more complete and structured form than the initial concept build. The current submission focuses on Levels 1 and 2 only, which together form a six-night campaign segment. This scope was selected to provide a clear vertical slice of the intended game while keeping the project manageable for the current stage of development.

The core gameplay loop is organized into three phases. During the day, the player explores the environment, collects resources, follows objectives, and prepares for the next attack. At night, the player must survive waves of zombies under increasing pressure. At dawn, the player receives points and rewards, then uses the shop to purchase weapons, armor, ammunition, healing items, and upgrades before continuing the run. This structure gives the prototype a clear rhythm and helps connect exploration, combat, and progression into one consistent experience.

The main design goal for this deliverable was to make preparation matter. In earlier versions, the daytime phase risked feeling like a simple break between combat segments. In the current build, player choices during the day have more visible consequences at night through crafting, contested drops, mission rewards, and resource management. At the same time, the project places greater emphasis on narrative context through radio transmissions, environmental storytelling, and mission-based progression tied to EVAC Command, Project Lazarus, Dr. Chen, and Subject 23.



(a) Level 1: Town Center. This area introduces the main objective route, early landmarks, and the first three day-night cycles.



(b) Level 2: Suburban district. This area expands the route, increases pressure, and continues the campaign.

Figure 1: Playable Deliverable 2 route. The current prototype intentionally focuses on Levels 1 and 2, while later areas are reserved for future deliverables.

2 Level Design and Player Guidance

The level design for Deliverable 2 aims to guide the player clearly while still allowing exploration. Rather than relying only on direct tutorial text, the prototype uses route structure, landmarks, mission prompts, and pacing changes to communicate where the player should go and what they should do next.

Level 1 serves as the introduction to the playable campaign. Its route moves the player through major landmarks such as the crash site, the military checkpoint, and the hospital. These locations are intentionally distinct so that the player can build a mental map of the area and understand the mission flow without becoming lost. Level 2 continues the same design approach, but the spaces are broader and more exposed. This creates a stronger sense of danger and makes navigation, scavenging, and defensive planning more demanding.

The two levels also support the learning curve of the prototype. Level 1 teaches the structure of exploration, preparation, and survival. Level 2 then asks the player to apply that understanding under greater pressure. This progression is important because it lets the prototype demonstrate not only basic functionality, but also how the game scales over time.

Player guidance is delivered through several overlapping cues:

- mission titles and live objective status through the in-game objective HUD,
- world-space objective markers that point toward the next important location,
- radio transmissions that give short narrative and tactical direction,
- recognizable landmarks that help the player build mental maps of each space,
- pacing shifts between day exploration and night defense.

This combination supports a balance between freedom and clarity. The player can search for resources, optional lore, and useful routes, but the game still signals the next critical task. This is especially important in a survival project, because difficulty should come from decision-making and pressure rather than uncertainty about the objective.

3 Narrative Design and World-Building

The narrative design follows a play-first approach. Story information is delivered during movement, scavenging, defense, and mission completion instead of being placed mainly in long cutscenes. This helps the prototype maintain pacing while still giving the player a clear sense of setting, purpose, and escalation.

In the Deliverable 2 build, the player is presented as the survivor of a failed evacuation attempt. Across six nights, the player receives instructions and updates from EVAC Command while moving through two unsafe districts in search of supplies, answers, and a path to extraction. As the campaign progresses, the radio dialogue and mission structure reveal more about Project Lazarus,

Dr. Chen, and Subject 23. This gives the run a stronger narrative arc than a simple wave-based survival mode.

We use several layers of narrative delivery:

- **Radio story beats** communicate the broad situation, major discoveries, and rising danger.
- **Story objectives** tie important plot information to specific spaces the player must reach.
- **Lore entries** add optional depth through logs, journals, military orders, and intercepted transmissions.
- **Environmental framing** makes each district feel like a place that was defended, abandoned, and partially erased.
- **Intro and ending sequences** give the playable run a stronger beginning and end-state.

This layered approach improves immersion because the main story remains understandable even if the player only follows core objectives, while optional exploration still provides additional depth. It also helps the prototype feel more unified. The player's goals, the spaces they move through, and the systems they interact with all support the same survival narrative.

4 Systems Design and Game Balance

Systems design in Deliverable 2 focuses on the relationship between daytime preparation and nighttime survival. The prototype uses a resource economy built around scavenged materials, blueprint tokens, points, health, armor, ammunition, and weapon access. Each part of this economy is meant to support a simple question for the player: what is worth spending now, and what should be saved for the next phase?

The daytime preparation layer is built around four visible goals:

- **Crafting meaningful support** such as ammo, healing, or combat modifiers.
- **Securing contested drops** for higher-value rewards under time pressure.
- **Completing mission leads** for points, ammo, and stronger narrative direction.
- **Looting efficiently** without ignoring the coming night.

These daytime systems directly affect the night phase. Strong preparation can give the player better weapons, more healing, stronger positioning, and a more stable start to the defense. Poor preparation can leave the player under-equipped and force riskier decisions once enemy pressure increases. This cause-and-effect relationship makes the balancing model easier to understand because the player can see how earlier choices influence later survival.

Balancing in this prototype is not based only on enemy strength. It is also based on how much information and opportunity the player receives before the attack begins. By connecting rewards,

mission flow, resource access, and danger, the game encourages planning rather than random trial and error.

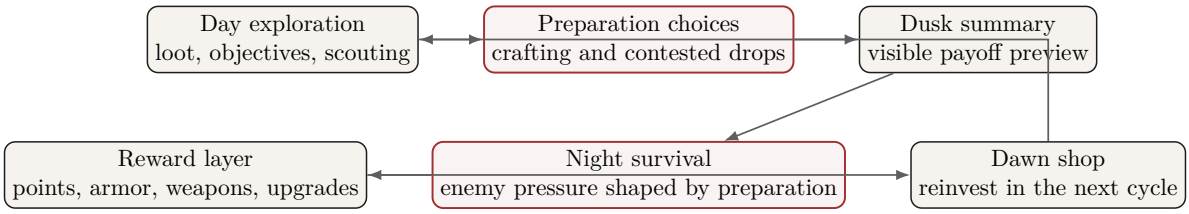


Figure 2: Core Deliverable 2 risk-reward loop. Daytime choices are designed to produce visible nighttime consequences, which then feed into rewards and future preparation.

5 Progression Systems and Rewards

Progression in the Deliverable 2 prototype works at two levels. The short-term loop is the day-night-dawn cycle within a single mission segment. The long-term loop is the full six-night run across Levels 1 and 2. Together, these systems give the player both immediate rewards and a broader sense of forward movement.

Short-term progression is visible every day. The player gathers resources, earns points, completes objectives, and reaches the dawn shop with a new set of choices. Long-term progression appears across the full prototype route. Later nights introduce stronger pressure, broader weapon availability, new objectives, and a transition from the Town Center to the Suburban district. This makes the run feel like a structured campaign rather than a repeating arena session.

Visible rewards are intentionally mixed:

- **Intrinsic rewards:** better routing, stronger survival discipline, and improved mastery of the game's systems.
- **Extrinsic rewards:** points, weapon access, armor, mission rewards, contested drop bundles, and upgrade purchases.
- **Narrative rewards:** logs, mission context, and story discoveries that make exploration feel worthwhile even when no direct combat reward is attached.

Keeping Level 2 in the prototype is important for this reason. It allows the player to carry momentum from one district into the next, observe how earlier spending and preparation affect later outcomes, and experience a clear transition in both setting and challenge. This greatly improves the report's progression and reward component because the prototype demonstrates campaign continuity rather than only a single-level sample.

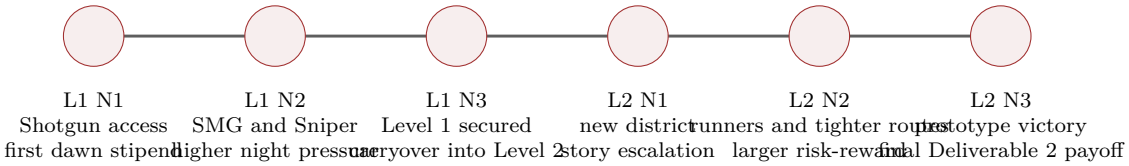


Figure 3: Six-night progression arc used in the Deliverable 2 build. The second level is important because it gives the prototype a visible campaign structure rather than a single-level slice.

6 Testing and Iteration

The prototype was improved through both observation-based feedback and automated runtime validation. Earlier testing suggested that the overall premise was understandable, but that some of the specific story details and system links were not always clear to players on first contact. In response, the design was revised to make communication more direct.

Several changes came from this process. Radio dialogue was strengthened so that the player received more direct context about the situation and the current objective. Mission framing was improved so that important tasks were easier to read. On the systems side, pickup hints, contested drop states, reward messaging, and dusk summaries were refined so that the player could better understand what had been gained during the day and why it mattered at night.

Systems iteration focused on clarity and fairness. The goal was not only to make mechanics functional, but also to ensure that players could understand them quickly and respond to them with informed decisions. This is especially important in a prototype, where the player must be able to recognize the intended design value of each system without needing extensive explanation outside the game.

Validation Area	Evidence Included in the Current Prototype
Crafting rules	Day-only crafting, resource costs, and per-day caps are covered by the PlayMode suite.
Contested events	Secure and expire flows are validated so the contested drop feature behaves predictably.
Resource clarity	Pickup purpose hints and reward messaging were added to reduce confusion.
Balance response	No-prep penalties and night escalation were checked to confirm that the prototype communicates cause and effect.
Regression safety	Existing regular crates and emergency night drops were retained and rechecked.
Iteration evidence	Narrative playtest notes and the latest local PlayMode results support the report's testing claims.

Table 1: Testing and iteration summary for Deliverable 2. The latest local PlayMode report records six passing tests for the current systems-focused suite.

7 Technical Notes

Several Unity systems supported this iteration:

- **Game flow and campaign state:** `GameManager` and `GameFlowController` coordinate the day, night, dawn, and level transition structure.
- **Mission guidance:** `StoryObjective`, the objective HUD, and objective markers provide in-world direction without relying on a heavy tutorial layer.
- **Narrative support:** radio updates, mission scripting, and lore systems support layered story delivery across the run.
- **Balance and preparation systems:** `CraftingSystem`, `SupplyCrate`, `Pickup`, and the wave balancing path connect day preparation to night pressure.
- **Submission structure:** the Deliverable 2 package is organized around the Unity project, the standalone build output, and this written PDF report.

8 Conclusion

Deliverable 2 represents a clear step forward from a basic survival prototype to a more cohesive game experience. The current build demonstrates readable level flow, stronger player guidance, clearer narrative integration, more understandable systems balancing, and a progression structure that benefits from spanning two linked levels instead of one isolated map.

Most importantly, the project's mechanics now support the setting and the intended player experience. The player is not only surviving enemy waves. They are exploring for resources, following mission leads, reacting to radio updates, and seeing how preparation, pressure, and reward connect across the campaign. This integration is the main achievement of the Deliverable 2 build and provides a strong foundation for future deliverables.